Single-Domain Generalization: A Scattering Characteristic-Driven Cross-Attention Model for Oriented SAR Object Detection

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***Abstract*—Oriented object detection in synthetic aperture radar (SAR) imagery often struggles to generalize to unknown domains, due to severe distribution shifts. Single-domain generalization (single-DG) aims to tackle this challenge by learning robust representations from only one source domain. However, existing single-DG methods typically fail to capture the essential features of rotating objects, as they neglect the unique physical properties introduced by the SAR imaging process. To address this, this article proposes the scattering characteristic-driven cross-attention model (SCCAM), which introduces a heuristic scattering representation to approximate the potential physical structures of SAR objects. Specifically, a pair of homologous images is constructed by perturbing the extremely low- and high-frequency components of the input SAR image, ensuring that they share domain-invariant information. The heuristic attribute scattering center features are then extracted to guide a bidirectional cross-attention module, enabling progressive interactions between homologous features. This design effectively mitigates feature distribution discrepancies, thereby facilitating the learning of domain-invariant representations. Experimental results on six single-DG benchmarks demonstrate that the proposed SCCAM significantly outperforms multiple classic and state-of-the-art DG models in oriented object detection across unknown SAR domains.**

***Index Terms*—Attribute scattering center (ASC), cross-attention, domain generation (DG), oriented object detection, synthetic aperture radar (SAR).**

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I. Introduction

ORIENTED object detection in synthetic aperture radar (SAR) data has become a pivotal technique in Earth observation, owing to its ability to precisely delineate geometry contours, orientations, and spatial positions of rotating objects. However, its effectiveness is often hindered by domain shifts arising from variations in geographic regions, sensor types, or imaging parameters. Such shifts introduce substantial discrepancies between training and testing distributions, violating the assumption of independent and identically distributed (i.i.d.) data. As a result, the performance of oriented object detectors tends to degrade severely when deployed in previously unseen environments. Unsupervised domain adaptation (UDA) methods attempt to mitigate domain shifts by exploiting unlabeled target domain data during training [1]. While UDA requires repeated access to target domains and is sensitive to imbalances in data volume between source and target domains. These characteristics limit its practicality and robustness in SAR applications, where target domains are typically unavailable in advance, and large-scale annotated data are difficult to obtain. To tackle these challenges, domain generalization (DG) has emerged as a promising paradigm that relieves domain shifts by training exclusively on available source domains [2]. Ideally, the resulting model achieves robust detection of rotating objects across arbitrary unseen SAR domains.

Recent years have witnessed notable advancements in DG research, which can be broadly categorized into three mainstream paradigms: data manipulation (including data augmentation [3] and data generation [4]); representation learning (including domain-invariant representation learning [5] and feature disentanglement [6]); and learning strategy (including meta-learning [7], self-supervised learning [8], and ensemble learning [9]). However, applying DG to oriented object detection in SAR imagery presents two key challenges. First, most DG methods assume access to multiple labeled source domains to learn generalized representations. While in the context of SAR imagery, the visual content is inherently difficult to interpret, making collecting well-annotated data across domains both labor-intensive and costly. This limitation has sparked interest in a more practical yet challenging setting: single-domain generalization (single-DG), where training is conducted using only one labeled source domain. Second, existing DG approaches

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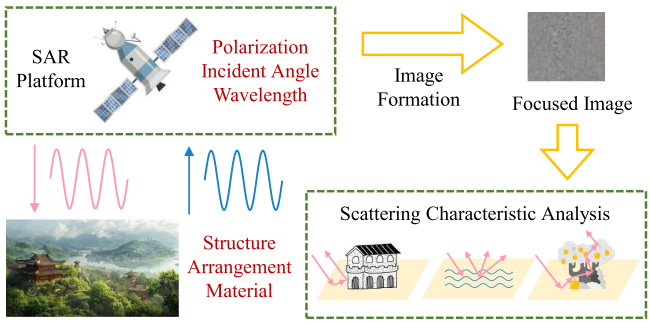


Fig. 1. Illustration of the physical principles of SAR imaging.

are predominantly designed for detection tasks using horizontal bounding boxes, which are insufficient for accurately detecting rotating objects in unseen scenes. These challenges underscore the need for a single-DG framework specifically tailored to oriented object detection in SAR imagery.

To this end, we make the first attempt to explore single-DG research for oriented SAR objects. Unlike natural images, SAR images are formed by microwave scattering from ground targets, typically lacking clear object boundaries, rich textures, and intuitive visual cues [10], [11]. Under significant domain shifts, both content representations and rotation angle prediction may be severely degraded. Notably, the physical properties inherent in SAR imagery (e.g., structure layout, geometry arrangement, and material composition) are strongly correlated with the electromagnetic scattering behavior of observed objects [12], as shown in Fig. 1. Such scattering characteristics are closely tied to object structures and are less sensitive to domain-specific appearance variations, making them reliable physical descriptors for single-DG scenarios. Inspired by this insight, we propose a scattering characteristic-driven cross-attention model (SCCAM). This model employs a heuristic scattering representation to approximate the underlying physical structures of SAR objects, thereby guiding the learning of domain-invariant representations.

In our approach, we leverage an observation from [5], [13] that the extremely low- and high-frequency components of images typically contain most domain-specific information. By randomly perturbing these frequency components, a homologous image that shares the domain-invariant information with the original input is generated for the subsequent generalization process. In addition, we incorporate the attribute scattering center (ASC) model [12], a parametric representation of electromagnetic scattering behavior. This model can effectively characterize the geometry structures and orientation cues of rotating objects, the heuristic ASC features are thus derived to guide the construction of the bidirectional cross-attention module. The domain-invariant representations are progressively captured through interactions between homologous image pairs. Extensive experiments on six single-DG benchmarks unequivocally verify that the proposed SCCAM achieves superior performance in detecting oriented objects on unknown SAR images, compared to classic and state-of-the-art DG models.

The main contributions are summarized as follows:

1) To the best of our knowledge, this work presents the first single-DG framework tailored for oriented SAR object detection, which exploits scattering characteristics to generalize across unknown SAR domains.
2) A pair of homologous images is constructed by perturbing domain-specific frequency components, thereby enriching domain diversity within a single source domain.
3) Driven by heuristic ASC features, a bidirectional cross-attention module is designed to align feature distributions between homologous image pairs, facilitating the extraction of domain-invariant knowledge.

The rest of this article is organized as follows. Section II reviews related works. Section III minutely describes our developed single-DG model, SCCAM. Section IV presents experimental results along with performance analysis. Finally, Section V concludes this article.

II. Related Works

A. Oriented Object Detection

In recent decades, oriented object detection has advanced considerably, benefiting from its ability to accurately estimate rotation angles and spatial positions. Xie et al. [14] proposed a general oriented detection framework by combining an oriented region proposal network with an oriented regions of interests (RoIs). Rather than directly regressing the vertices of the bounding box, Xu et al. [15] described multioriented objects using relative gliding offsets along the horizontal border. Liu et al. [16] introduced a circle anchor net to fundamentally circumvent the regression ambiguity in traditional horizontal anchor-based methods. Dong et al. [17] dynamically adjusted convolution kernel sampling positions to adaptively capture deformation features in dense industrial scenes, effectively enhancing robustness to shape and scale variations. In the SAR domain, Qiu et al. [18] integrated staged circular angle modeling and angle-aware sample assignment to enhance rotation prediction, thereby improving the detection performance in densely distributed ships. Sun et al. [19] investigated the elliptical scattering properties of ships, and proposed a Gaussian probability distribution loss to mitigate regression imbalance across targets of varying scales and orientations. Despite these advancements, oriented object detection in unseen domains remains challenging due to the widespread domain shifts in real-world scenarios.

B. Cross-Domain Transfer Learning Methods

Cross-domain transfer learning aims to enhance model generalization under domain shifts by transferring knowledge learned from a source domain. A representative research direction is UDA, which aligns feature distributions between source and target domains through domain-alignment [1], domain-mapping [20], or self-straining strategies [21]. However, UDA methods rely on access to unlabeled or weakly labeled target-domain data during training, which is impractical in real-world applications. To relax this requirement, DG methods learn domain-invariant representations from multiple labeled source domains, enabling generalization to unseen target domains. In

SAR imagery, collecting annotated data across multiple domains is particularly challenging due to complex imaging mechanisms and high annotation costs. This limitation naturally motivates the study of single-SD.

C. Single-DG Object Detection

Motivated by the practical constraints of cross-domain data collection, single-DG object detection has recently emerged as a promising solution for addressing out-of-distribution (OOD) challenges in detection tasks. To improve generalization ability, Danish et al. [22] aligned classification and localization outputs across multiple views. Rui et al. [4] introduced a progressive diversity generation strategy to synthesize images in a fully unsupervised manner. Cui et al. [23] designed a selective physical information fusion module that incorporates the wavelet transform into the neural network to strengthen domain-invariant features. Inspired by optical perturbations arising from the light–atmosphere interactions, Xu et al. [24] applied the image frequency spectrum to simulate real-world variations. In the context of SAR ship detection, Yuan et al. [6] combined low-order latent space filling with high-order instance disentanglement. However, these methods typically rely on horizontal detection boxes, struggling to reliably predict orientations under domain shifts. To overcome this, Bi et al. [25] pioneered a generalized oriented object detector by incorporating rotation-aware content consistency learning and style consistency learning.

Despite recent progress, research on rotating objects remains in its infancy. This underscores the urgency for more accurate single-DG models tailored to SAR imagery. Noteworthily, electromagnetic scattering characteristics play a key role in interpreting SAR images [12], offering valuable inspiration for generalizing oriented object detectors to unseen domains.

D. Scattering Characteristic-Guided Method

Motivated by the scattering behavior described by the geometrical theory of diffraction (GTD), scattering characteristics provide a compact and physically meaningful signal representation for SAR targets [26], [27]. Bounding upon this foundation, a variety of scattering characteristic-guided methods have been extensively explored in recent years. Observing the stable performance of scattering features under extended operating conditions, Zhang et al. [28] integrated scattering center features with deep CNN representations. Peng et al. [29] introduced an adversarial attack algorithm guided by scattering models, generating adversarial perturbations in the form of electromagnetic scattering response. Fang et al. [30] utilized diverse frequency time modulation to conceal its original scattering signature, effectively camouflaging SAR targets. Huang et al. [31] presented a scattering enhancement and feature fusion network to highlight the scattering points of aircraft targets. Li et al. [32] introduced a zero-shot recognition model that integrates scattering characteristics-guided semantic embedding generation and polarization characteristics-guided distributional correction. He et al. [33] applied a joint representation of visual and scattering features to enhance ship detection in complex-valued SAR images. Collectively, these studies highlight the effectiveness and potential of scattering characteristics in enhancing the interpretability of SAR imagery.

III. Proposed SCCAM Method

In the single-DG work, we have access to a source domain $D_s = \{(x_j^s, y_j^s, b_j^s)\}_{j=1}^{N_s}$, whose parameters x, y, and b represent the images, labels, and bounding boxes annotations, separately. The objective is to learn an oriented object detector that generalizes effectively from a labeled source domain D_s to one or more unseen SAR domains $D_t = \{x_k^t\}_{k=1}^{N_t}$. Notably, D_s and D_t are sampled from distinct data distributions while sharing an identical label space.

Significant domain shifts often hinder both visual content representation and rotation angle estimation in an oriented object detector. To address this issue, this work concentrates on robustly characterizing rotating SAR objects in a domain-invariant manner. It is worthwhile mentioning that scattering characteristics have proven effective in describing physical structures and rotation directions [12]. Inspired by this, we propose a novel SCCAM, tailored for the single-DG scenario. The approach leverages a heuristic scattering representation to approximate the physical structures of rotating SAR objects, thereby guiding the extraction of domain-invariant features. The overall framework is illustrated in Fig. 2, primarily consisting of following two key procedures.

1) Homologous Image Pair Construction

The domain-specific frequency components f_{LH} of the source domain D_s are randomly perturbed to generate a homologous domain $D_{s'}$. Each image pair shares domain-invariant information while differing in domain-specific attributes, providing diverse inputs for the subsequent generalization process.

2) ASC-Driven Cross-Attention

The heuristic ASC features (i.e., F_{a1} and F_{a2}) are extracted as supervisory signals and integrated with multiscale spatial features (i.e., F_{s1} and F_{s2}) to build a bidirectional cross-attention module. Through interactions between homologous image pairs, domain-invariant features (i.e., $F_{\text{invraiant1}}$ and $F_{\text{invariant2}}$) are progressively captured. After RoIAlign, the consistency of feature distribution between rotating instances (i.e., F_{ins1} and F_{ins2}) is further enforced.

A. Homologous Image Pair Construction

Compared to DG research in multiple domains, single-DG inherently suffers from limited and less diverse domain knowledge. To alleviate this constraint, homologous image pairs are constructed to enrich intradomain diversity meanwhile preserving task-relevant structural content [34], [35]. Prior studies have verified that frequency spectrum serves as an effective means to disentangle image components, i.e., the extremely low- and high-frequency bands (denoted as f_{LH}) characterize domain-specific information, while the remaining ones (denoted as f_M) encode domain-invariant content [5], [13]. In the context of

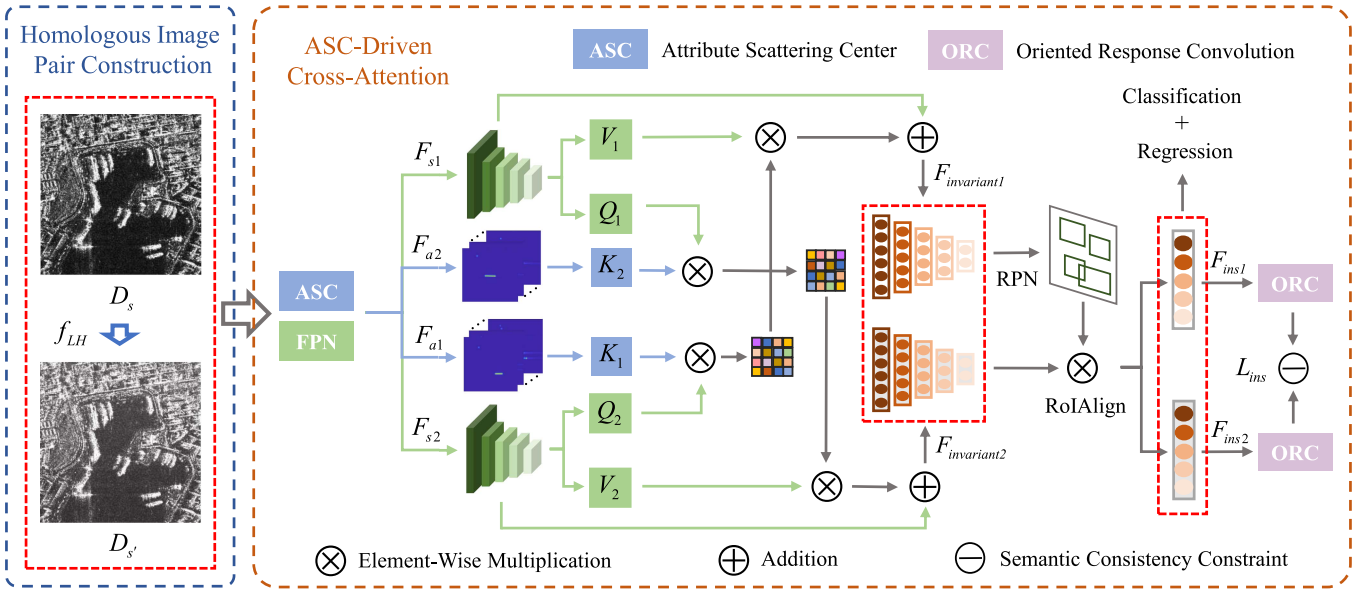


Fig. 2. Framework of the proposed Single-DG model, termed the SCCAM. First, a homologous domain $D_{s'}$ is constructed by randomly transforming the extremely low- and high-frequency components f_{LH} of the source domain D_s. Next, the heuristic ASC features F_{a1}, F_{a2}, and multiscale spatial features F_{s1}, F_{s2} are separately extracted from the homologous image pairs. They construct the bidirectional cross-attention module to progressively capture domain-invariant features $F_{\text{invariant1}}$, $F_{\text{invariant2}}$. Following RoIAlign, the feature distributions between rotating instances F_{ins1} and F_{ins2} are further aligned.

SAR imagery, the f_{LH} is more sensitive to speckle noise and clutter related to sensor types or observation scenes, whereas the f_M predominantly captures object structures and scattering characteristics that are invariant across domains [36]. Building on this principle, the homologous domain $D_{s'}$ is synthesized by preserving f_M and randomly perturbing f_{LH} according to a Gaussian distribution.

Specifically, for an input SAR image $x \in \mathbb{R}^{H\times W}$, its frequency spectrum $\mathscr{F}(x)$ is obtained through the discrete Cosine transform $\mathscr{F}(\cdot)$. A soft band-pass filter $\mathcal{M}(r)$ is defined to decompose $\mathscr{F}(x)$ into domain-specific frequency components f_{LH} and domain-invariant frequency components f_M, that is,

$$f_{LH} = \mathcal{M}(r)\cdot\mathscr{F}(x), \quad f_M = (1-\mathcal{M}(r))\cdot\mathscr{F}(x) \tag{1}$$

$$\mathcal{M}(r) = 1 - \tau r + \tau\cdot \mathrm{ReLU}(r-r_1) + \mathrm{ReLU}(r-r_2) \tag{2}$$

where $\tau = (1-r_1)/r_1, \mathrm{ReLU}(x) = \max(0,x)$ denotes the rectified linear unit. r presents the normalized radial distance of each frequency point. Although optical and SAR imagery differ fundamentally in sensing mechanisms, their spatial frequency spectra share similar structural regularities [37]. Therefore, the thresholds $r_1 = 0.01$ and $r_2 = 0.7$ follow the configuration in [13], respectively, define the boundaries between low–medium and medium–high frequencies. Subsequently, the homologous frequency spectrum is synthesized by randomizing f_{LH} with a Gaussian distribution. The corresponding homologous image is reconstructed via the Inverse discrete Cosine transform $\mathscr{F}'(\cdot)$, defined as

$$x' = \mathscr{F}'\left(f_{LH}\cdot(1+\mathcal{N}(0,1)) + f_M\right). \tag{3}$$

Noteworthily, the homologous image pairs constructed based on frequency perturbation are not intended to explicitly simulate complex real-world domain shifts, but rather to diversify domain-specific variations while preserving domain-invariant content. As illustrated in Fig. 3, the visualized construction pipeline provides intuitive evidence for the effectiveness of the proposed homologous image pairs, revealing how different frequency components contribute to domain-specific and domain-invariant representations. Specifically, this construction process yields several interesting observations: 1) images reconstructed solely from f_M components retain structural details from the source domain D_s, validating their ability to capture domain-invariant information; 2) images derived from f_{LH} components exhibit weak semantic structures but noticeable intensity fluctuations, indicating that they primarily characterize domain-specific content; 3) homologous images $D_{s'}$ synthesized by combining f_M with randomized f_{LH}, simulate diverse environmental conditions while preserving original object structures. Such image pairs effectively enrich data diversity and encourage model adapting to a wider range of domain variations, thereby benefiting the downstream generalization.

B. ASC-Driven Cross-Attention

In practical scenes, the location and orientation predicted by oriented object detectors are often degraded by complex and diverse domain shifts. An efficient solution is to focus on domain-invariant features that robustly represent rotating SAR objects, while suppressing domain-specific interference. It is noteworthy that scattering characteristics provide physically meaningful cues about object structures and rotation directions [12]. Leveraging this property, the heuristic ASC features are

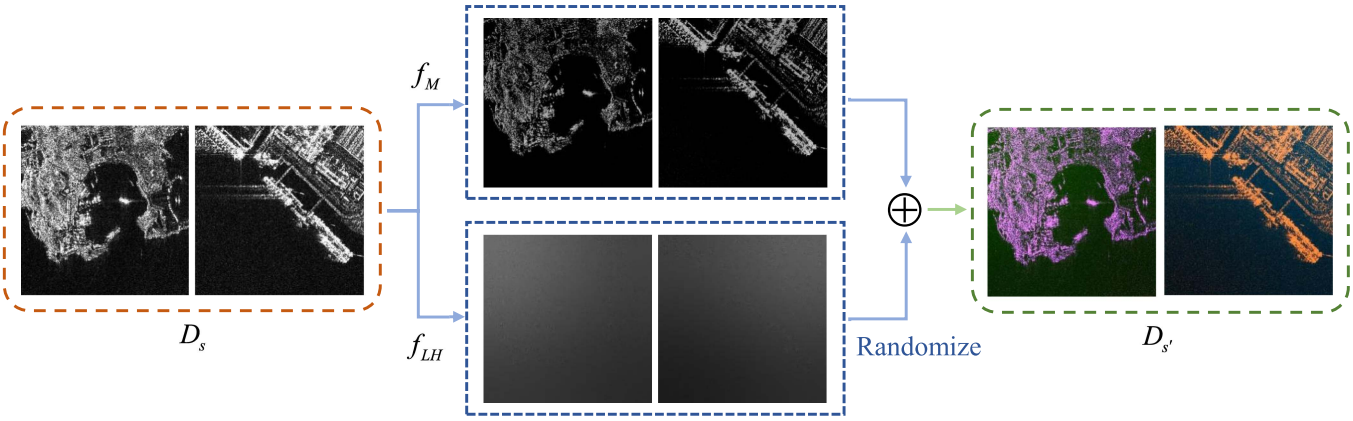


Fig. 3. Illustration of the homologous image pair construction. The source domain D_s is decomposed into extremely low- and high-frequency components f_{LH} containing domain-specific information, and the remaining frequency components f_M characterizing domain-invariant information. The homologous domain $D_{s'}$ is synthesized by integrating the original f_M with a randomized version of f_{LH}.

extracted as supervision signals, guiding the bidirectional cross-attention module to refine multiscale spatial features. Through interactions between homologous image pairs, shared feature components are progressively distilled as domain-invariant knowledge. Furthermore, oriented response convolution (ORC) is known for capturing global/local rotations for orientation estimation [38]. In our framework, ORC is applied to build an oriented classifier that reduces feature distribution disparities between rotating instances. The thorough methodology is described in the following sections.

1) Heuristic ASC Feature Extraction: Primarily, to effectively characterize the underlying physical properties of SAR images, the ASC model is employed. As established in GTD [26], [27], the total backscattered field in the high-frequency regions can be approximated by the superposition of responses from individual scattering centers. Based on this principle, the ASC features are parameterized as

$$E(f,\varphi;\Theta_I)=\sum_{i=1}^{I}E_i(f,\varphi;\theta_i) \tag{4}$$

where f and φ, respectively, denote the operating frequency and aspect angle of the radar. The parameter set for I individual scatters is defined as

$$\Theta_I=\{\theta_i|\theta_i=[A_i,x_i,y_i,\alpha_i,\gamma_i,L_i,\overline{\varphi}_i],1\le i\le I\} \tag{5}$$

where A_i is the amplitude of the echo, and (x_i,y_i) are the spatial coordinates of the ith scatter. The parameter α_i models the frequency and geometry dependence, while γ_i accounts for aspect dependence. In addition, L_i and $\overline{\varphi}_i$ represent the effective length and orientation angle of the distributed scattering center, respectively. Distinct scattering structures are characterized by unique parameter configurations Θ_I. The typical configurations observed in real-world SAR scenarios are summarized in Table I.

Particularly, for SAR objects observed in actual circumstances, the computational cost of employing a typical genetic algorithm (GA) [39] for parameter optimization is prohibitively high. To this end, we design a simple yet physically grounded heuristic scattering feature modeling strategy. A set of predefined physical templates $\{\theta_p^{\text{temp}}\}_{p=1}^{P}$ is constructed according to scattering types listed in Table I. A fixed number of scattering centers is randomly sampled from the top 20% high-intensity regions of the SAR image, ensuring strong correlations with actual target structures. For the ith scattering center, the parameter vector θ_i is computed by combining one randomly selected template $\theta_{p(i)}^{\text{temp}}$ with a bounded perturbation $\Delta\theta_i$, that is,

$$\theta_i=\theta_{p(i)}^{\text{temp}}+\Delta\theta_i. \tag{6}$$

Since $\theta_{p(i)}^{\text{temp}}$ reflects real-world scattering behaviors, this design preserves the physical plausibility of the ASC model, while introducing controlled variability. The electromagnetic response $E_i(f,\varphi;\theta_i)$ of the $i-th$ scatter is then calculated as

$$\begin{aligned}E_i(f,\varphi;\theta_i)=&A_i\cdot\left(\frac{f}{f_c}\right)^{\alpha_i}\\&\cdot\exp\left(-\mathrm{j}\frac{4\pi f}{c}(x_i\cos\varphi+y_i\sin\varphi)\right)\\&\cdot\operatorname{sinc}\left(\frac{2\pi f}{c}L_i\sin(\varphi-\bar{\varphi}_i)\right)\cdot\exp\left(-2\pi f y_i\sin\varphi\right)\end{aligned} \tag{7}$$

where f_c and c separately denote the radar central frequency and propagation speed of light. The extraction process of heuristic ASC features is demonstrated in Fig. 4(b). Initially, the

TABLE I
GEOMETRY SCATTERING TYPES DIFFERENTIATED BY PARAMETERS SETTING

	Geometric Scattering Type	α	γ	L	$\overline{\varphi}$
Localized	Trihedral	1	>0	0	0
	Top Hat	0.5			
	Sphere	0			
	Corner Diffraction	-1			
Distributed	Dihedral	1	0	>0	$\neq 0$
	Cylinder	0.5			
	Edge Broadside	0			
	Edge Diffraction	-0.5			

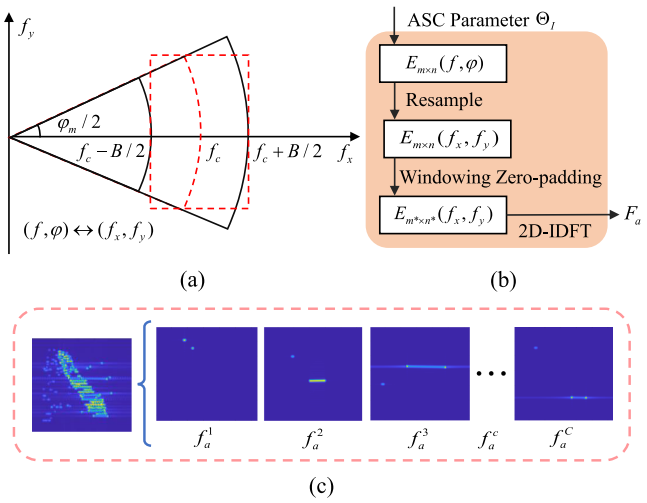


Fig. 4. Illustration of the heuristic ASC feature generation. (a) Resampling of polar format data onto a rectangular grid in the Cartesian plane. (b) Extraction process. (c) Resulting multi-channel feature maps $F_a = \{F_a^c\}_{c=1}^C$.

total backscattered response in the polar plane is denoted as $E_{m\times n}(f, \varphi)$. As illustrated in Fig. 4(a), it is formed by sampling m frequency points and n aspect angle points over $f \in [f_c - B/2, f_c + B/2]$, $\varphi \in [-\varphi_m/2, \varphi_m/2]$. Where B represents the radar bandwidth and φ_m denotes the aperture accumulation angle. Subsequently, the polar data is interpolated onto a uniform Cartesian grid (f_x, f_y) using the coordinate transformation

$$f_x = f\cos(\varphi), \quad f_y = f\sin(\varphi). \tag{8}$$

Under this mapping, $E_{m\times n}(f_x, f_y)$ is uniformly sampled over $f_x \in [f_c - B/2, f_c + B/2]$ and $f_y \in [-f_c \cdot \sin(\varphi_m/2), f_c \cdot \sin(\varphi_m/2)]$, as shown in the rectangle region of Fig. 4(a). To ensure high-quality image formation, a window function is applied to suppress sidelobes, followed by zero-padding to $m^* \times n^*$ samples. Finally, a 2-D inverse discrete Fourier transform (2D-IDFT) yields scattering response maps in the spatial domain. In our approach, the resulting heuristic ASC features $F_a = \{F_a^c\}_{c=1}^C$ are derived from multichannel scattering response maps, where $C = 256$. As illustrated in Fig. 4(c), each channel complementarily encodes physical attributes of the target, e.g., amplitude decay, geometry dependence, and aspect variation. These physics-driven features reflect the underlying SAR scattering mechanisms, thereby enhancing the robustness of the oriented object detector against spatial and domain variations.

Particularly, since the F_a are derived from predefined physical templates $\{\theta_p^{\text{temp}}\}_{p=1}^P$, their generalization may be constrained when encountering OOD target domains, such as new composite materials or complex imaging geometries. To mitigate potential degradation in such scenarios, the F_a are further refined through learnable linear projections that adaptively adjust the heuristic representations during training. This lightweight refinement improves the adaptability and robustness of F_a to unseen scattering variations.

2) Bidirectional Cross-Attention: Generally speaking, observed SAR objects vary with gauging distances and imaging mechanisms. To capture such variations, multiscale spatial features $F_s = \{F_s^l\}_{l=1}^5$ are extracted using a feature pyramid network (FPN) [41], describing visual details from coarse to fine granularity. For each homologous image pair with distinct domain-specific content, the corresponding multiscale spatial features (i.e., F_{s1} and F_{s2}) tend to emphasize diverse visual components. Deeper alignment is thus required to uncover domain-invariant cues. Cross-attention mechanisms enable features from one source to query and selectively integrate information from another [42]. In our single-DG framework, a bidirectional cross-attention module is introduced to enable mutual feature interaction between the homologous image pair. Unlike one-way alignment, the bidirectional design allows semantic cues to be exchanged in both directions, facilitating shared representation learning while suppressing domain-specific interference. Crucially, this process is guided by heuristic ASC features, which encode physically meaningful scattering characteristics. These physical priors direct attention toward structurally stable and domain-invariant features, ensuring robust detection even under significant domain shifts.

To illustrate the bidirectional cross-attention process, we consider the lth scale as an example. Given heuristic ASC features (i.e., F_{a1} and F_{a2}) and the corresponding spatial features (i.e., F_{s1}^l and F_{s2}^l), a parallel structure with average pooling and max-pooling is applied. The query (Q) and value (V) tensors are generated from the spatial features, while the key (K) tensors are derived from the heuristic ASC features, as follows:

$$\begin{aligned} Q_\xi, V_\xi &= 0.5 \cdot \text{avgPool}(F_{s\xi}^l) + 0.5 \cdot \text{maxPool}(F_{s\xi}^l) \\ K_\xi &= 0.5 \cdot \text{avgPool}(F_{a\xi}) + 0.5 \cdot \text{maxPool}(F_{a\xi}) \end{aligned} \tag{9}$$

where $\xi := \{1, 2\}$ indexes the homologous image pair. Taking F_{s1}^l as an example, an attention score matrix is computed by the dot product between Q_2 and K_1, followed by a Softmax operation to produce the attention weights. These weights aggregate V_1 to align with its homologous counterpart. A residual connection then adds the original F_{s1}^l, yielding the refined domain-invariant features, that is,

$$F_{\text{invariant1}}^l = \text{Softmax}\left(\frac{Q_2 K_1^T}{\sqrt{d}}\right) V_1 + F_{s1}^l \tag{10}$$

where d is the dimensionality of Q and K vectors, $\sqrt{d}$ is used to scale the dot product and stabilize the gradient. The same operation is symmetrically applied to F_{s2}^l to ensure mutual enhancement across the image pairs, defined as follows:

$$F_{\text{invariant2}}^l = \text{Softmax}\left(\frac{Q_1 K_2^T}{\sqrt{d}}\right) V_2 + F_{s2}^l. \tag{11}$$

By applying the above procedure across all pyramid levels, the multiscale spatial features F_{s1} and F_{s2} yield domain-invariant features $F_{\text{invariant1}} = \{F_{\text{invariant1}}^l\}_{l=1}^5$ and $F_{\text{invariant2}} = \{F_{\text{invariant2}}^l\}_{l=1}^5$. To explicitly demonstrate their effectiveness, Fig. 5 visualizes the extracted domain-invariant features of homologous image pairs. The first row shows three representative image pairs from distinct scenes, and the second row

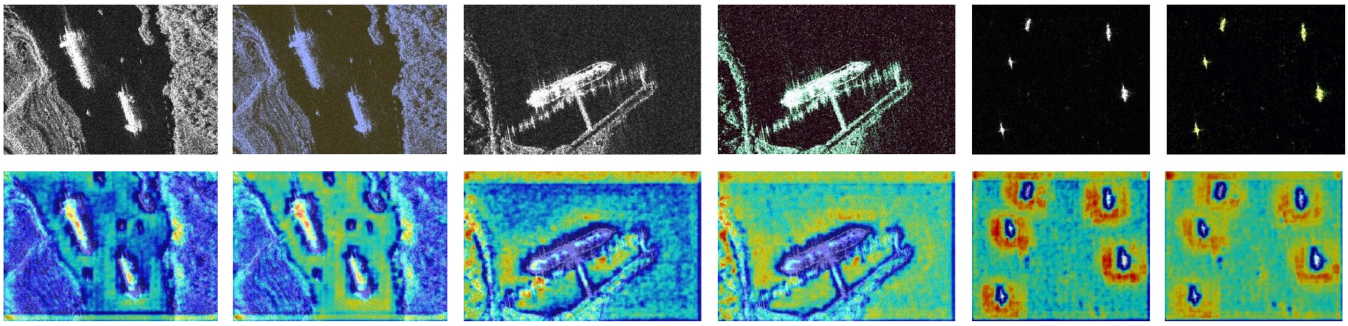


Fig. 5. Illustration of extracted domain-invariant features. [Row1]: Three homologous image pairs from the SSDD dataset [40], covering varying scene complexities. [Row2]: Heat maps of the corresponding domain-invariant feature maps.

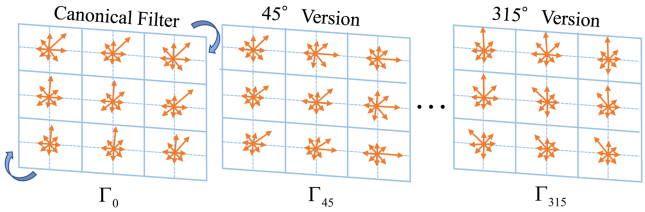


Fig. 6. Illustration of an ARF bank. Each ARF actively performs 7 rotations during convolution, producing 8-directional responses on a 3×3 grid. Starting from the canonical filter Γ_0, the kernel is clockwise rotated 7 directions, generating $\Gamma_{45}, \Gamma_{90}, \cdots, \Gamma_{315}$.

displays the corresponding heat maps. Obviously, the responses are strongly concentrated on ship targets, with prominent activations along structural edges. That aligns with the physical expectations, as strong scattering responses typically occur at geometric and electromagnetic discontinuities. Moreover, $F_{\text{invariant1}}$ and $F_{\text{invariant2}}$ present only minor variations, both consistently emphasizing rotating SAR objects. These observations suggest that the extracted domain-invariant features are closely related to physically meaningful scattering behaviors and effectively capture geometric structures under varying scene conditions.

3) Consistency Enhancement for Rotating Instances: To further refine domain-invariant features under rotational variations, we enhance the consistency of feature distributions between rotating instances. Following RoIAlign, a homologous pair of instance features (F_{ins1} and F_{ins2}) is generated. To incorporate orientation awareness, an ORC that models receptive field responses from multiple directions is introduced to construct oriented classifiers. Building upon this, the feature distributions of F_{ins1} and F_{ins2} are progressively aligned.

Specifically, the ORC consists of a bank of active rotating filters (ARFs), as demonstrated in Fig. 6. Each ARF is defined with size $K \times K \times N$, where N is the number of orientations. Starting from a canonical filter Γ_0, $N-1$ additional filters Γ_{θ_m} are generated by clockwise rotations with $\theta_m = 2\pi m/N$ for $m = 0, 1, \ldots, N-1$. This produces an ARF bank of dimensions $N \times K \times K \times N$, encoding directional responses across orientations. For each ARF, the feature map is convolved across its N orientation channels to produce rotation-sensitive outputs. Leveraging all N ARFs, the oriented classifier is defined as

$$\Gamma(F_{ins\xi}) = \sum_{m=0}^{N-1} \sum_{n=0}^{N-1} \Gamma_{\theta_m}^{(n)} * F_{ins\xi}^{(n)} \tag{12}$$

where $\xi := \{1, 2\}$ indexes the homologous instance pair. $\Gamma_{\theta_m}^{(n)}$ and $F_{ins\xi}^{(n)}$ represent the nth orientation channel of Γ_{θ_m} and $F_{\text{ins}\xi}$. Unless otherwise specified, N is determined as 8.

To enhance semantic consistency across the homologous instance pair, an oriented instance constraint loss is defined as follows:

$$L_{\text{ins}} = \mathbb{E}(\Gamma(F_{\text{ins1}}), y^s) + \mathbb{E}(\Gamma(F_{\text{ins2}}), y^s) \tag{13}$$

where $\mathbb{E}$ is the cross-entropy loss function, y^s is the ground-truth label from the source domain D_s. This loss drives F_{ins1} and F_{ins2} to be mapped into a shared semantic space, thereby aligning their feature distributions under a unified label supervision. This alignment further strengthens the domain-invariance of instance-level representations.

Ultimately, the proposed SCCAM optimizes a combined objective function integrating two fundamental detection losses with the computed L_{ins}, with the definition as

$$L_{\text{SCCAM}} = L_{\text{cls}} + L_{\text{reg}} + \beta L_{\text{ins}} \tag{14}$$

where L_{cls} denotes the classification loss for category prediction, while L_{reg} represents the regression loss for precise localization. The hyperparameter β balances the contribution of these loss components in the training objective.

IV. Experimental Results and Analysis

A. Datasets and Evaluation Criteria

To evaluate the generalization capability of the proposed SCCAM, extensive experiments are conducted on three representative SAR datasets: HRSID [43], SSDD [40], and RSDD-SAR [44] datasets. A brief description of each dataset is provided as follows.

1) *HRSID* [43]: This dataset comprises 16 951 ship instances derived from TerraSAR-X, Sentinel-1 and TanDEM-X sensors, distributed across 5604 SAR images of size 800×800. They are split into 3642 training images and

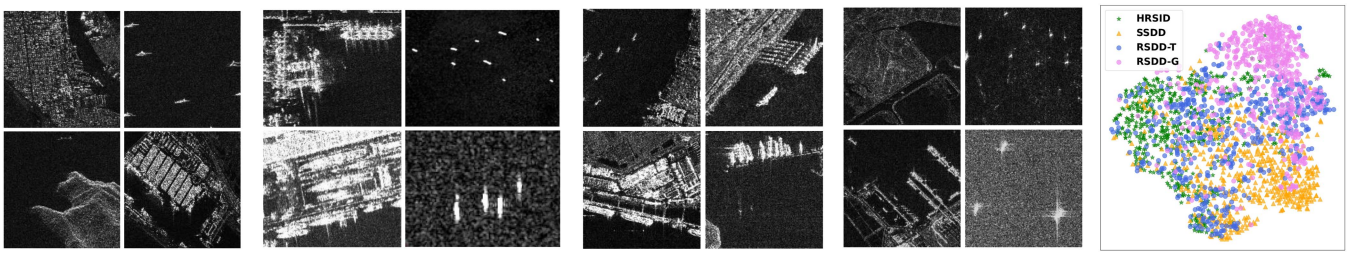


Fig. 7. Illustration of SAR images, selected from (a) HRSID [43]; (b) SSDD [40]; (c) RSDD-T [44]; (d) RSDD-G [44]. The corresponding t-SNE visualization, derived from features extracted by the Gliding Vertex [15] model, is demonstrated in (e).

TABLE II
IMAGING PARAMETERS OF THE EMPLOYED SAR DATASETS

Datasets	Imaging Sensor	Polarization Mode	No. of Train	No. of Test	Resolution	Size
HRSID [43]	TerraSAR-X, Sentinel-1, TanDEM-X	HH, VV, HV	3642	1962	$0.5m, 1.0m, 3.0m$	800×800
SSDD [40]	RadarSat-2, TerraSAR-X, Sentinel-1	HH, VV, VH, HV	928	232	$1m - 15m$	190-668
RSDD-T [44]	TerraSAR-X	HH, VV	986	385	$2m - 20m$	512×512
RSDD-G [44]	Gaofen-3	HH, VV, VH, HV, DH, DV	4014	1615	$2m - 20m$	512×512

1962 testing images. Notably, the oriented bounding box annotations are adapted from semantic segmentation annotations.

2) *SSDD* [40]: This dataset contains 1160 SAR images acquired from RadarSat-2, TerraSAR-X, and Sentinel-1 sensors, with image sizes vary from 190 to 668 pixels. They include 2456 ship instances spanning both inshore and offshore scenes, partitioned into 928 training images and 232 testing images.
3) *RSDD-SAR* [44]: This dataset offers 7000 SAR images originating from Gaofen-3 and TerraSAR-X sensors, encompassing 10263 ship instances. They are divided into 5000 training images and 2000 testing images, with a size of 512×512. This dataset features diverse imaging configurations, including observation modes, polarization types, and resolutions.

Noteworthily, SAR datasets differ significantly due to variations in sensor types, which operate under diverse imaging parameters, platform altitudes, polarization modes, and surface coverage conditions. These factors contribute to considerable domain diversities, primarily manifested as the scattering characteristics of rotating objects, imaging resolution, noise levels, and background structures. Particularly, the RSDD-SAR dataset includes images separately captured by the Gaofen-3 and TerraSAR-X sensors, exhibiting distinct scattering patterns and texture details. For experimental purposes, we divide this dataset into two subsets: RSDD-T (images from TerraSAR-X) and RSDD-G (images from Gaofen-3). Examples of the four SAR datasets are demonstrated in Fig. 7(a)–(d), with their imaging parameters summarized in Table II. In addition, Fig. 7(e) presents their t-SNE visualization, based on features extracted from the baseline detector. Some overlap exists among the data distributions, reflecting the general robustness of the trained detector. However, features from the Gaofen-3 sensors form a distinct cluster, separated from the others due to their unique radar bands, polarization modes, and noise characteristics. Accordingly, two complementary single-DG settings are considered in our experiments. First, the RSDD-G dataset is treated as an unknown target domain, while HRSID, SSDD, and RSDD-T are each used as a single source domain for training, to evaluate cross-DG under different SAR imaging conditions. Second, RSDD-G is used as the sole source domain and generalized to HRSID, SSDD, and RSDD-T, evaluating robustness across multiple unseen SAR domains.

To assess the detection accuracy for a single object category (i.e., ships), four commonly-used criteria are adopted, including Precision rate, Recall rate, average precision (AP), and F1-score. Precision and recall separately measure the false alarm rate and missed detection rate of the oriented detector, while AP and F1-score provide a more comprehensive evaluation by balancing these two aspects.

B. Implementation Details

Two classic oriented object detection models (i.e., Oriented RCNN [14] and Gliding Vertex [15]) are adopted as baseline detectors, both employing ResNet-101 [51] for feature extraction. All experiments are implemented in the PyTorch framework with stochastic gradient descent (SGD) optimization, where the weight decay is 0.0001 and the momentum is 0.9. The model is trained for 24 epochs, starting with an initial learning rate of 0.005, which is reduced by a factor of 10 at epochs 16 and 22. During the testing phase, generalization performance is assessed by the four aforementioned metrics under an intersection over union threshold of 0.5. In addition, the hyperparameter β in (14) is empirically tuned for each baseline detector, where $\beta = 0.75$ for oriented RCNN and $\beta = 0.25$ for gliding vertex.

TABLE III
RESULTS (%) OF SINGLE-DG TASKS FROM THREE SOURCE DOMAINS, BASED ON THE ORIENTED RCNN [14] AND GLIDING VERTEX [15] MODELS

	Method	Oriented RCNN				Gliding Vertex				Avg. Perf.	
		AP	Recall	Precision	F1-score	AP	Recall	Precision	F1-score	AP	F1-score
RSDD-T → RSDD-G	Baseline	89.09	91.32	85.42	88.27	88.85	90.44	84.26	87.24	88.97	87.76
	IBN-Net [3]	**89.72**	92.25	87.89	**90.02**	**89.13**	90.35	**88.18**	**89.25**	**89.43**	**89.63**
	ISW [45]	88.52	91.14	83.01	86.89	87.96	90.67	78.35	84.06	88.24	85.47
	CDSD [46]	88.56	90.63	81.38	85.76	88.66	90.63	64.20	75.16	88.61	80.46
	MLRT [47]	89.20	**92.34**	81.72	86.71	88.00	**91.60**	69.31	78.91	88.60	82.81
	OA-DG [48]	89.62	91.83	**88.03**	**89.89**	**89.13**	90.95	83.27	86.94	89.38	88.42
	DivAlign [22]	89.24	90.90	**88.04**	89.45	88.71	90.95	76.95	83.37	88.98	86. 41
	PhysAug [24]	89.56	92.02	87.47	89.69	89.05	90.77	**84.82**	**87.69**	89.31	**88.69**
	SRCD [49]	**89.76**	**92.85**	85.59	89.07	89.06	90.81	84.03	87.29	**89.41**	88.18
	Boost [50]	87.51	90.72	67.04	77.10	87.95	90.70	67.97	77.71	87.73	77.40
	Yuan [6]	88.64	92.30	37.58	53.41	89.05	**91.28**	27.39	42.14	88.85	47.77
	GOOD [25]	89.09	92.06	67.83	78.11	88.16	90.95	67.05	77.19	88.63	77.65
	SCCAM [Ours]	**89.85**	**92.71**	**89.16**	**90.90**	**89.41**	**91.65**	**84.22**	**87.78**	**89.63**	**89.34**
SSDD → RSDD-G	Baseline	67.44	79.35	**77.90**	**78.62**	67.23	77.77	62.12	69.07	67.34	**73.84**
	IBN-Net [3]	72.16	80.65	69.72	74.79	70.83	81.11	54.88	65.47	71.50	70.13
	ISW [45]	65.63	79.21	65.98	71.99	65.24	76.46	**73.63**	**75.02**	65.44	73.51
	CDSD [46]	**74.86**	**82.27**	73.56	**77.67**	**74.09**	**81.53**	53.90	64.90	**74.48**	71.28
	MLRT [47]	65.07	78.24	73.60	75.85	65.94	80.10	62.42	70.16	65.51	73.01
	OA-DG [48]	**76.85**	**83.25**	68.76	75.31	71.16	80.28	59.61	68.42	**74.01**	71.87
	DivAlign [22]	73.01	80.00	71.33	75.42	73.65	80.60	55.58	65.79	73.33	70.60
	PhysAug [24]	73.77	80.14	**74.76**	77.36	71.34	80.23	**69.94**	**74.73**	72.56	**76.04**
	SRCD [49]	72.26	81.44	64.88	72.22	72.35	80.23	63.35	70.80	72.31	71.51
	Boost [50]	62.18	76.19	21.60	33.66	64.70	77.54	56.47	65.35	63.44	49.50
	Yuan [6]	65.71	78.79	47.21	59.04	**74.81**	81.11	27.58	41.16	70.26	50.10
	GOOD [25]	74.39	81.76	69.37	75.06	73.10	**81.76**	40.82	54.45	73.75	64.76
	SCCAM [Ours]	**78.43**	**87.47**	**73.63**	**79.96**	**75.94**	**83.67**	**63.60**	**72.27**	**77.19**	**76.11**
HRSID → RSDD-G	Baseline	78.87	82.51	**77.88**	80.13	78.04	82.97	63.77	72.11	78.46	76.12
	IBN-Net [3]	**80.41**	88.03	72.96	79.79	79.35	87.66	70.22	77.98	**79.88**	78.88
	ISW [45]	79.65	85.94	73.26	79.10	**79.56**	85.38	**81.85**	**83.58**	79.61	**81.34**
	CDSD [46]	79.58	85.15	**77.56**	**81.18**	78.87	83.76	**73.43**	78.26	79.23	79.72
	MLRT [47]	79.76	88.72	72.04	79.51	79.49	**88.07**	65.70	75.26	79.63	77.39
	OA-DG [48]	**80.10**	85.80	69.91	77.04	**79.65**	**87.89**	59.67	71.08	**79.88**	74.06
	DivAlign [22]	79.71	**88.86**	56.82	69.32	79.06	86.59	51.83	64.85	79.39	67.08
	PhysAug [24]	79.84	86.50	76.77	**81.35**	79.31	87.61	72.70	**79.46**	79.58	**80.40**
	SRCD [49]	80.02	**88.91**	74.12	80.84	79.46	86.13	**74.06**	**79.64**	79.74	**80.24**
	Boost [50]	75.17	82.74	32.53	46.70	76.52	84.18	47.00	60.32	75.85	53.51
	Yuan [6]	78.67	84.92	31.17	45.60	78.53	84.22	25.89	39.61	78.60	42.60
	GOOD [25]	79.34	84.87	**82.05**	**83.44**	78.58	85.89	42.46	56.83	78.96	70.13
	SCCAM [Ours]	**82.75**	**90.26**	64.70	75.37	**80.03**	**89.74**	58.73	71.00	**81.39**	73.18

C. Performance Comparison and Analysis

In this subsection, the generalization performance of the proposed SCCAM is evaluated against several classic and state-of-the-art DG approaches, using oriented RCNN [14] and gliding vertex [15] as the baseline frameworks. All experiments are conducted under consistent testing conditions to ensure a fair comparison. These compared methods include those originally designed for natural images (i.e., IBN-Net [3], ISW [45], CDSD [46], MLRT [47], OA-DG [48], DivAlign [22], PhysAug [24], SRCD [49], and Boost [50]), SAR images (i.e., Yuan [6]), and oriented objects (i.e., GOOD [25]). The source codes are publicly available from their respective authors/websites. The subsequent analysis emphasizes the performance gains achieved by the proposed SCCAM over these models.

1) Performance Analysis of Single-DG to Single Unseen Domain: To compare the generalization performance, we first conduct single-DG tasks from multiple source domains (i.e., RSDD-T, SSDD, and HRSID) to a target domain (i.e., RSDD-G), as shown in Table III. For clearer visualization, the top-three results in terms of each evaluation metric (i.e., AP, Recall, Precision, and F1-score) are highlighted in bold, with red, blue, and black colors. Notably, the proposed SCCAM ranks the top-three performance 20 times, indicating its superior capability to detect rotating objects from unseen SAR domains. Moreover, Table III also records the average performance across the two baseline detectors. On the RSDD-G dataset, most DG models (including the baselines) already demonstrate strong generalization capabilities. Such a high baseline makes further improvement particularly challenging. In this context, the

TABLE IV
RESULTS (%) OF SINGLE-DG TASKS TO THREE TARGET DOMAINS, BASED ON THE ORIENTED RCNN [14] AND GLIDING VERTEX [15] MODELS

	Method	RSDD-G → RSDD-T				RSDD-G → SSDD				RSDD-G → HRSID				Avg. Perf.	
		AP	Recall	Precision	F1-score	AP	Recall	Precision	F1-score	AP	Recall	Precision	F1-score	AP	F1-score
Oriented RCNN	Baseline	70.01	79.81	47.08	59.22	56.83	67.58	45.95	54.70	58.64	63.79	**54.37**	**58.70**	61.83	57.54
	IBN-Net [3]	69.31	75.15	**68.89**	**71.88**	57.56	64.10	**74.63**	**68.97**	**59.56**	60.34	**72.30**	**65.78**	62.14	**68.88**
	ISW [45]	70.05	72.04	**79.97**	**75.80**	56.71	64.10	**64.46**	**64.28**	**60.07**	60.02	**73.49**	**66.08**	62.28	**68.72**
	CDSD [46]	68.05	77.30	34.92	48.11	56.18	67.22	18.11	28.53	58.00	65.29	36.37	46.72	60.74	41.12
	MLRT [47]	73.13	80.29	39.00	52.50	58.01	67.03	**53.90**	**59.75**	59.51	**65.53**	45.51	53.72	63.55	55.32
	OA-DG [48]	**76.22**	**81.36**	52.10	63.52	56.91	66.67	44.99	53.73	58.80	65.28	46.66	54.42	63.98	57.22
	DivAlign [22]	69.79	78.26	44.92	57.08	57.58	68.32	46.86	55.59	59.27	63.94	49.68	55.92	62.21	56.19
	PhysAug [24]	**75.55**	**81.60**	47.10	59.73	**61.08**	**72.34**	35.08	47.25	59.07	64.50	49.20	55.82	**65.23**	54.26
	SRCD [49]	75.28	80.29	50.79	62.22	**61.82**	**70.33**	38.95	50.13	59.24	65.06	51.65	57.58	**65.45**	56.65
	Boost [50]	65.96	75.03	38.62	50.99	52.03	52.45	37.48	43.72	54.93	60.88	44.87	51.66	57.64	48.79
	Yuan [6]	68.93	78.73	43.55	56.08	54.61	67.22	45.54	54.30	58.16	**66.64**	49.37	56.72	60.57	55.70
	GOOD [25]	72.40	80.29	35.82	49.54	53.64	67.22	36.44	47.26	58.55	63.25	46.86	53.84	61.53	50.21
	SCCAM [Ours]	**76.85**	**81.96**	**52.41**	**63.94**	**64.17**	**75.09**	41.16	53.17	**59.84**	**67.05**	47.45	55.57	**66.95**	**57.56**
Gliding Vertex	Baseline	66.93	75.51	32.81	45.74	42.67	54.40	19.39	28.59	56.37	64.40	29.48	40.45	55.32	38.26
	IBN-Net [3]	67.89	70.37	**71.48**	**70.92**	54.81	62.45	**53.28**	**57.50**	59.01	62.39	**53.82**	**57.79**	60.57	**62.07**
	ISW [45]	69.30	75.27	**67.38**	**71.11**	**55.76**	61.90	**62.71**	**62.30**	**59.47**	63.75	**56.76**	**60.05**	61.51	**64.49**
	CDSD [46]	68.23	78.61	27.64	40.90	50.80	**65.20**	17.06	27.04	57.34	**65.46**	26.28	37.50	58.79	35.15
	MLRT [47]	**69.78**	79.09	50.08	61.33	55.04	65.02	41.75	50.85	59.02	64.89	48.17	55.29	61.28	55.82
	OA-DG [48]	**73.59**	**80.17**	43.32	56.25	54.24	65.12	36.97	47.16	58.51	65.24	45.87	53.87	**62.11**	52.43
	DivAlign [22]	69.10	78.85	29.85	43.31	54.70	**67.03**	24.16	35.52	57.17	64.33	31.67	42.44	60.32	40.42
	PhysAug [24]	69.25	**79.57**	50.53	61.81	**56.38**	**69.05**	29.20	41.04	59.03	64.94	40.17	49.64	**61.55**	50.83
	SRCD [49]	69.72	77.54	**52.81**	**62.83**	54.40	65.13	35.03	45.56	58.52	65.14	48.57	55.65	60.88	54.68
	Boost [50]	65.52	72.40	38.97	50.67	43.51	50.55	29.87	37.55	55.51	61.32	43.39	50.82	54.85	46.35
	Yuan [6]	67.70	75.15	29.78	42.66	48.88	62.27	28.75	39.34	56.58	62.52	28.37	39.03	57.72	40.34
	GOOD [25]	68.88	79.21	39.77	52.95	52.38	63.37	34.81	44.94	**59.34**	**67.25**	27.40	38.94	60.20	45.61
	SCCAM [Ours]	**75.16**	**80.76**	49.06	61.04	**55.12**	**65.20**	**41.93**	**51.04**	**59.12**	**67.52**	**48.67**	**56.57**	**63.13**	**56.21**

proposed SCCAM achieves notable performance gains, securing the highest average AP and the highly competitive average F1-score. These results highlight its ability to effectively capture domain-invariant characteristics, thereby maintaining strong robustness and accuracy under domain shifts.

2) Performance Analysis of Single-DG to Multiple Unseen Domains: Table IV presents the generalization performance on the single-DG tasks from a source domain (i.e., RSDD-G) to multiple target domains (i.e., RSDD-T, SSDD, and HRSID). As in the previous analysis, the top-three performances are highlighted in bold, using red, blue, and black colors. In this case, the proposed SCCAM obtains the most top-three performance (i.e., 18 times), outperforming ISW (i.e., 15 times) and IBN-Net (i.e., 13 times). This demonstrates that the SCCAM consistently surpasses competing methods in generalizing across multiple unseen SAR domains. The variation in oriented detection results among the three target datasets also reveals the substantial domain shifts introduced by different imaging sensors. The corresponding average AP and F1-score are thus reported to comprehensively assess the cross-domain robustness. From which one can be observed that the proposed SCCAM consistently outperforms other DG methods on both baseline detectors, achieving the best average AP and third-best F1-score.

Particularly, the proposed SCCAM shows a clear advantage over the GOOD model, which is also dedicated to oriented objects. This gap largely stems from its reliance on optical images, whose data distributions diverge markedly from SAR images. Such discrepancies hinder GOOD to learn domain-invariant knowledge suitable for SAR-specific tasks. In contrast, SCCAM adequately considers the unique physical properties of SAR imaging, leveraging scattering characteristics to guide the feature interactions between homologous pairs. This design enables the progressive learning of domain-invariant representations, allowing SCCAM to generalize to a wide range of unknown SAR domains.

3) Visualization of Oriented Detection Results: Fig. 8 visualizes the generalization performance of the proposed SCCAM and several representative models, under the single-DG tasks (i.e., RSDD-G → RSDD-T/SSDD/HRSID). The compared models include the baseline detector (i.e., Oriented RCNN [14]), a DG model designed for natural images (i.e., PhysAug [24]), DG models tailored for rotating objects (i.e., GOOD [25] and the proposed SCCAM). The figure presents detection results on the RSDD-T dataset ($1st$ and $2nd$ rows), SSDD dataset ($3rd$ and $4th$ rows), and HRSID dataset ($5th$ and $6th$ rows). Ground truth is marked with green boxes, correct detections with blue boxes, and missed or false detections with red boxes. From Fig. 8, it can be observed that offshore targets generally achieve higher detection accuracy than inshore ones. A plausible reason is that the source domain collected by the Gaofen-3 sensor, predominantly consists of offshore scenes. They share similar imaging conditions and scattering characteristics with offshore targets, which partially alleviates domain shifts. In contrast, inshore scenes are characterized by higher scene complexity and

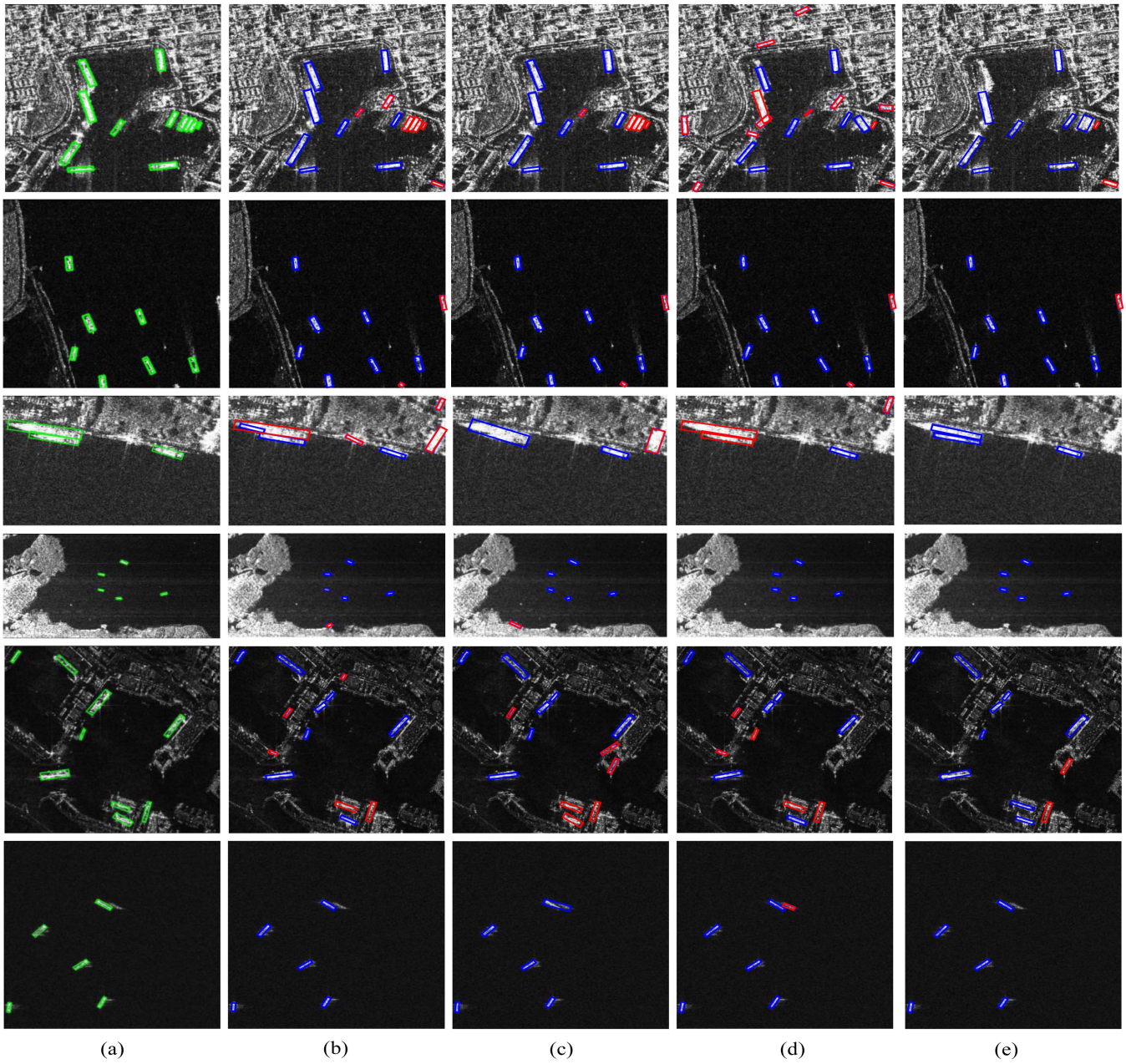


Fig. 8. Illustration of oriented detection results for single-DG models trained on the RSDD-G dataset. [Row 1–2]. (a) Original images selected from the RSDD-T dataset, and oriented detection results generated by; (b) Oriented RCNN; (c) PhysAug; (d) GOOD; (e) the proposed SCCAM. [Row 3–4]: corresponding results on the SSDD dataset. [Row 5–6]: corresponding results on the HRSID dataset. Green boxes indicate the ground truth of SAR objects, blue boxes present correct detections, and red boxes highlight missed or false detections.

stronger background interference, making it difficult for models to accurately capture target structures. Under such challenging conditions, the proposed SCCAM still effectively captures rotating ship targets, indicating its superior ability to extract robust and physically meaningful representations in complex SAR environments.

4) Ablation Study of Different Components: To assess the contribution of each component within the proposed SCCAM framework, a series of ablation studies are conducted on the single-DG task SSDD $\rightarrow$ RSDD-G. As shown in Table V, the examined components comprise: homologous image pair construction (denoted as HIPC), ASC-based guidance (denoted as ASC), and bidirectional cross-attention module (denoted as cross-attention). Since cross-attention relies on the presence of homologous image pairs, variants using cross-attention alone are thus excluded from the ablation analysis. Both oriented RCNN and Gliding Vertex are adopted as baseline detectors, with their average performance reported for a comprehensive comparison. For each evaluation metric (i.e., AP, recall, precision, and F1-score), the best results are highlighted in bold. From Table V, we observe that each component brings noticeable improvements over the baselines, confirming their effectiveness in enhancing

TABLE V
ABLATION RESULTS (%) OF SINGLE-DG ON SSDD → RSDD-G

Module			Oriented RCNN [14]				Gliding Vertex [15]				Avg. Perf.	
HIPC	ASC	Cross-Attention	AP	Recall	Precision	F1-score	AP	Recall	Precision	F1-score	AP	F1-score
-	-	-	67.44	79.35	77.90	78.62	67.23	77.77	62.12	69.07	67.34	73.84
★	-	-	74.57	83.11	74.22	78.41	73.74	80.42	72.18	**76.08**	74.16	**77.25**
-	★	-	72.02	80.05	**79.82**	72.02	73.46	**84.22**	58.97	69.37	72.74	74.65
★	★	-	74.60	81.72	68.71	74.65	70.06	79.58	**72.27**	75.75	72.33	75.20
★	-	★	73.72	81.53	73.82	77.48	72.66	80.28	66.44	72.71	73.19	75.10
★	★	★	**78.43**	**87.47**	73.63	**79.96**	**75.94**	83.67	63.60	72.27	**77.19**	76.11

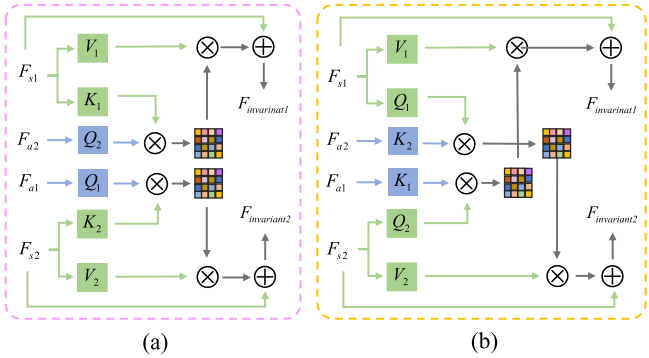


Fig. 9. Illustration of the bidirectional cross-attention module, where heuristic ASC features serve as (a) query tensors and (b) key tensors.

TABLE VI
PERFORMANCE COMPARISON (%) OF SINGLE-DG TASKS USING DIFFERENT BIDIRECTIONAL CROSS-ATTENTION STRATEGIES

	Criteria	Oriented RCNN [14]		Gliding Vertex [15]	
		ASC-Q	ASC-K [Ours]	ASC-Q	ASC-K [Ours]
RSDD-T	AP	75.26	76.85 (+1.59)	70.00	75.16 (+5.16)
	Recall	81.00	81.96 (+0.96)	79.81	80.76 (+0.95)
	Precision	48.88	52.41 (+3.53)	52.07	49.06 (-3.01)
	F1-score	60.97	63.94 (+2.97)	63.02	61.04 (-1.98)
SSDD	AP	55.96	64.17 (+8.21)	51.75	55.12 (+3.37)
	Recall	68.50	75.09 (+6.59)	61.54	65.20 (+3.66)
	Precision	42.31	41.16 (-1.15)	40.93	41.93 (+1.00)
	F1-score	52.31	53.17 (+0.86)	49.16	51.04 (+1.88)
HRSID	AP	59.34	59.84 (+0.5)	58.74	59.12 (+0.38)
	Recall	66.14	67.05 (+0.91)	65.97	67.52 (+1.55)
	Precision	47.57	47.45 (-0.12)	49.59	48.67 (-0.92)
	F1-score	55.34	55.57 (+0.23)	56.62	56.57 (-0.05)

generalization to unseen domains. Specifically, HIPC enriches the knowledge diversity within a single SAR domain, effectively contributing to detection robustness; ASC introduces heuristic guidance that characterizes geometry structures and rotation cues, assisting the model in capturing visual and physical properties. Interestingly, not all arbitrary combinations of components yield additive gains, which emphasizes the complementarity and rationality of the proposed integration strategy. The proposed SCCAM effectively combines ASC-based guidance with bidirectional cross-attention on homologous image pairs. This design significantly boosts the capability to detect previously unseen rotating objects, achieving the highest average AP and a relatively competitive average F1-score.

5) Performance Comparison on Bidirectional Cross-Attention: In this subsection, we further examine the impact of the bidirectional cross-attention module within the proposed SCCAM, by comparing two variants for single-DG tasks targeting three unseen domains. The first variant, illustrated in Fig. 9(a), adopts heuristic ASC features as query tensors and is referred to as ASC-Q. The second one applies heuristic ASC features as key tensors (denoted as ASC-K and utilized in the proposed SCCAM), as demonstrated in Fig. 9(b). Unlike ASC-Q [52], where both key and value tensors come solely from spatial features, ASC-K further facilitates cross-modal interactions between spatial features (i.e., V_1 and V_2) and heuristic ASC features (i.e., K_1 and K_2).

The experiments are conducted on both the oriented RCNN and gliding vertex models, as demonstrated in Table VI. The performance changes relative to the ASC-Q are highlighted, with improvements in red and declines in blue. It can be observed that both ASC-Q and ASC-K achieve favorable generalization across unseen target domains, validating the effectiveness of the proposed bidirectional cross-attention mechanism for DG. It facilitates the extraction of domain-invariant representations by enabling feature interactions between homologous image pairs, effectively mitigating overfitting to domain-specific patterns in single-DG scenarios. Notably, the proposed SCCAM employing ASC-K consistently outperforms its ASC-Q counterpart, achieving notable performance gains. This improvement arises from the stronger cross-modal interactions introduced by the ASC-K, where physical cues from heuristic ASC features are integrated into the learning process of spatial features. Such a design allows the module to dynamically attend to shared components across homologous image pairs, yielding more robust domain-invariant representations and consequently better generalization to previously unseen SAR domains.

6) Impact of Hyperparameter β: To investigate the contribution of oriented instance constraint loss in the training objective, we evaluate the performance of the proposed SCCAM under different values of hyperparameter β, ranging from $\beta = 0.125$ to $\beta = 1.000$. The corresponding results on Single-DG tasks RSDD → RSDD-T/SSDD/HRSID are reposted in Table VII, using oriented RCNN and gliding vertex as baseline detectors. The best performance in terms of AP is highlighted in bold for clarity. As observed, the proposed SCCAM consistently achieves favorable performance across a wide range of β values,

TABLE VII
PERFORMANCE COMPARISON (%) OF SINGLE-DG TASKS USING DIFFERENT PARAMETERS β, BASED ON ORIENTED RCNN [14] AND GLIDING VERTEX [15] MODELS

	Datasets	0.125	0.250	0.375	0.500	0.625	0.750	0.875	1.000
Oriented RCNN	RSDD-T	76.22	76.01	76.10	76.82	75.28	**76.85**	76.34	76.52
	SSDD	61.33	62.38	59.92	61.85	61.62	**64.17**	60.12	61.76
	HRSID	59.59	59.51	59.44	59.25	59.21	**59.84**	59.05	58.93
Gliding Vertex	RSDD-T	69.90	**75.16**	73.05	69.82	69.56	69.92	69.91	69.52
	SSDD	54.01	**55.12**	54.34	51.21	53.41	53.85	52.08	54.01
	HRSID	58.46	**59.12**	58.69	58.53	57.84	58.22	58.75	58.32

TABLE VIII
COMPUTATIONAL COMPLEXITY COMPARISON OF DIFFERENT MODELS BASED ON ORIENTED RCNN MODEL [14]

Method	FLOPs (G)	Parameters (M)	Per Time (ms)
Baseline	240.10	60.12	154.71
IBN-Net [3]	240.15	60.34	152.40
ISW [45]	263.49	62.52	147.34
CDSD [46]	240.10	60.12	214.23
MLRT [47]	240.10	60.12	266.31
OA-DG [48]	240.10	60.64	238.37
DivAlign [22]	240.10	60.12	145.83
PhysAug [24]	240.10	60.12	149.56
Yuan [6]	240.10	214.30	198.21
SRCD [49]	240.10	115.69	234.37
Boost [50]	1485.72	60.12	434.42
GOOD [25]	899.33	62.22	237.86
SCCAM [Ours]	466.35	61.28	934.09

with only moderate performance variation. This demonstrates that the model exhibits strong generalization ability across different SAR target domains and is not overly sensitive to the specific choice of β. Notably, optimal performance is obtained at $\beta = 0.75$ for the oriented RCNN and at $\beta = 0.25$ for the gliding vertex. Accordingly, these values are adopted as the default hyperparameter settings for (14) in the respective configurations.

7) Computational Complexity Analysis: In addition to detection accuracy, computational complexity is an indispensable factor for evaluating the practical applicability of the single-DG model. To this end, we conduct a comprehensive complexity analysis of all compared models based on the oriented RCNN model, for the single-DG task RSDD-G $\rightarrow$ SSDD. As summarized in Table VIII, the evaluation metrics include the floating-point operations (FLOPs), the number of parameters, and the average inference time per SAR image. All experiments were performed on a unified platform equipped with a single NVIDIA RTX 2080 Ti GPU. It can be observed that the SCCAM entails higher computational complexity than the baseline and most compared models. This additional cost primarily stems from the heuristic ASC feature extraction and the bidirectional cross-attention applied across multiple FPN levels. Nevertheless, the proposed SCCAM achieves a reasonable tradeoff between detection accuracy and complexity, as the increased computational overhead yields substantial gains in generalization to unseen SAR domains and enhances the interpretability of physically motivated features. Furthermore, the computational burden can be mitigated in future work through channel pruning or knowledge distillation, making the model more suitable for real-time or edge-device applications.

V. CONCLUSION

In this article, a single-DG model is proposed to generalize oriented object detection to unknown SAR domains, termed the SCCAM. To effectively capture the essential knowledge of rotating objects, heuristic scattering characteristics are extracted to approximate their underlying physical structures. Specifically, a pair of homologous images is constructed by randomizing domain-specific information in the original SAR images. The heuristic ASC features are then extracted from these images, guiding to bulid the bidirectional cross-attention module. By modeling dependencies between each image pair, their feature distribution discrepancies are progressively reduced, facilitating the extraction of domain-invariant representations. Experimental results on six single-DG benchmarks demonstrate that the proposed SCCAM significantly outperforms both classic and state-of-the-art models in detecting rotating objects from previously unseen SAR images. In future work, we aim to further enhance the efficiency and scalability of the proposed SCCAM to facilitate practical deployment in large-scale or real-time SAR applications. In addition, exploring more adaptive representations of physically meaningful scattering characteristics will contribute to improving the robustness and generalization capability in the single-DG setting.

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